

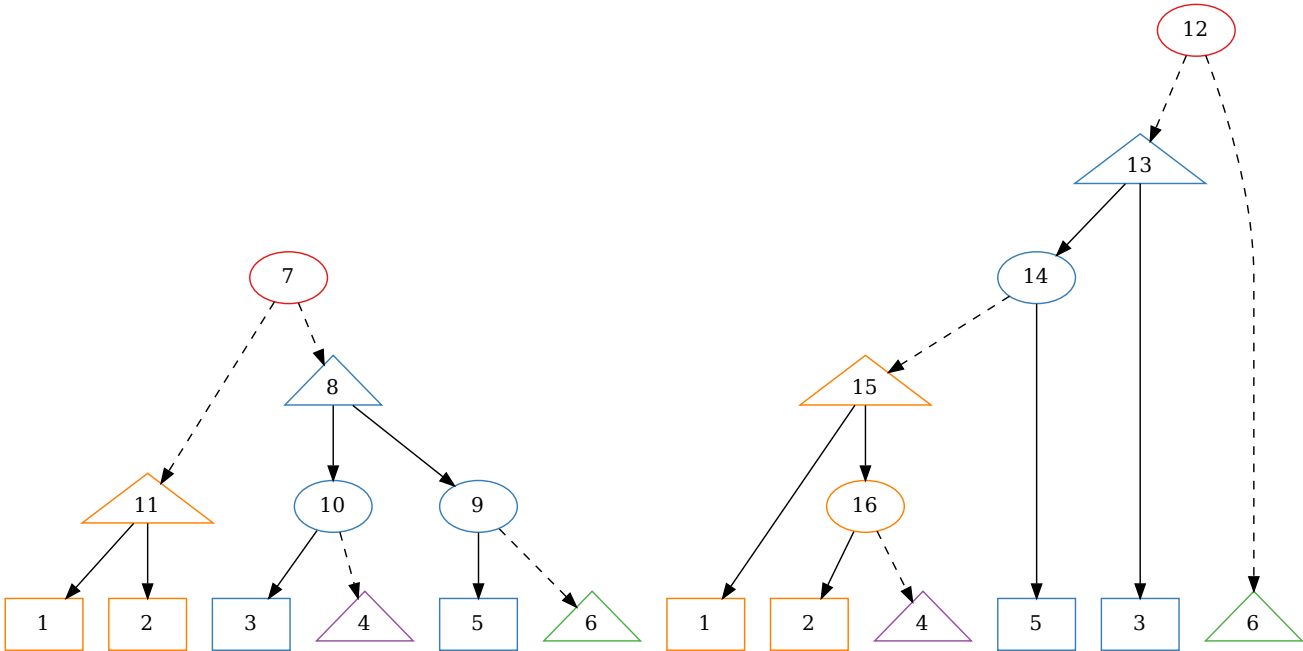
tiny01.nw

Instance:

```
#s hash "0x118494c254c36869026cfc"
#s name "tiny01"
#s desc "Example shown on https://pacechallenge.org"
#p 2 6
(((5,6),(3,4)),(1,2));
((((4,2),1),5),3),6);
#x treedecomp [2,[ [8,16], [8,11,16], [1,11,15], [2,11,16], [7,8,11], [8,10,16], [3,10,13], [4,10,16], [8,9], [5,9,14], [6,9,12]], [[1,2], [1,6], [1,9], [2,3], [2,4], [2,5], [6,7], [6,8], [9,10], [9,11]]]
```

Feasible solution:

```
(5,3);
6;
4;
(1,2);
```



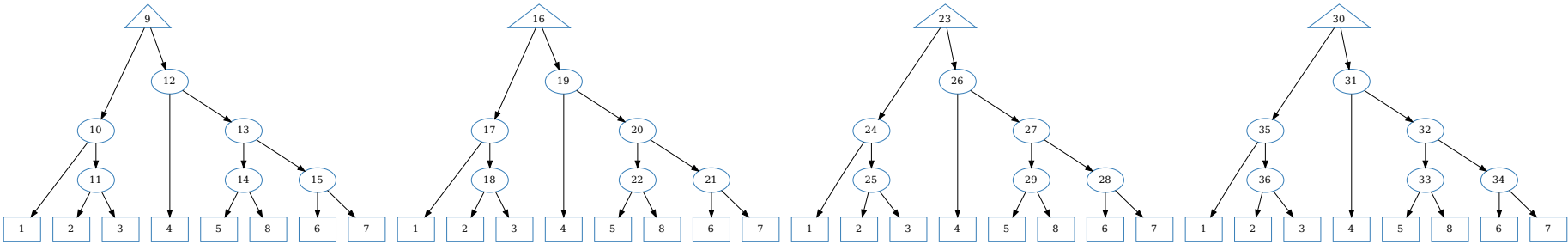
tiny02.nw

Instance:

```
#s hash "0x2f63d46fd664affc3a47f7"
#s name "tiny02"
#s desc "This instance consists of multiple representations of the same tree. Hence each tree is a MAF."
#p 4 8
((1,(2,3)),(((8,5),(6,7)),4));
(((3,2),1),(((6,7),(5,8)),4));
(((2,3),1),(4,(((6,7),(5,8)))));
((4,((8,5),(6,7))),((1,(3,2))));
#x treedecomposition [2,[[10],[1,10],[1,17],[1,24],[1,35],[10,11],[2,3,11],[2,3,18],[2,3,25],[2,3,36],[9,10,12],[4,12],[4,19],[4,26],[4,31],[12,13],[5,8],[5,8,14],[5,8,22],[5,8,29],[5,8,33],[6,7],[6,7,15],[6,7,21],[6,7,28],[6,7,34],[16],[20],[23],[27],[30],[32]],[[1,2],[1,6],[1,11],[1,17],[1,22],[1,27],[1,28],[1,29],[1,30],[1,31],[1,32],[2,3],[2,4],[2,5],[6,7],[7,8],[7,9],[7,10],[11,12],[11,16],[12,13],[12,14],[12,15],[17,18],[17,19],[17,20],[17,21],[22,23],[22,24],[22,25],[22,26]]]
```

Feasible solution:

```
((1,(2,3)),(((8,5),(6,7)),4));
```



tiny03.nw

Instance:

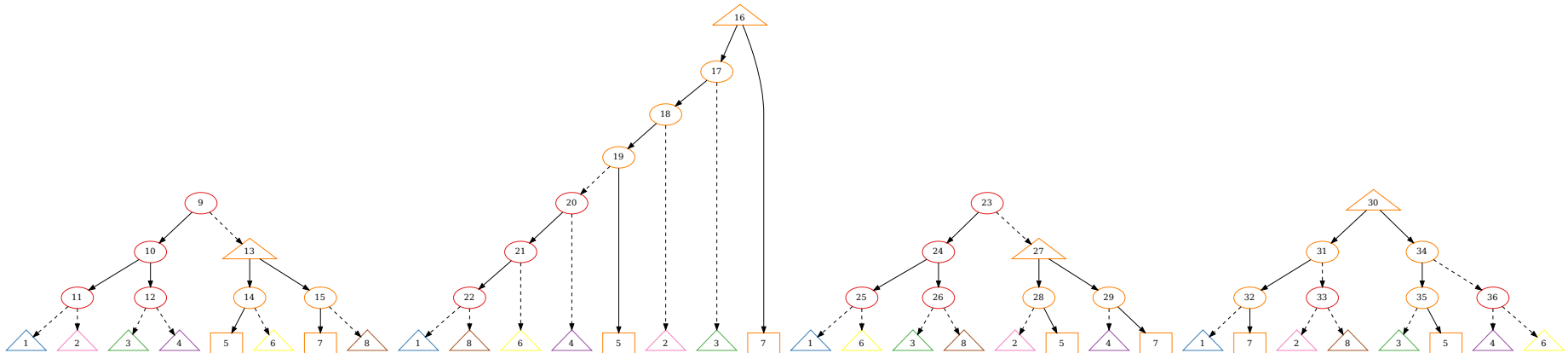
```

1 #s hash "0x2d0792e16776755723b31f"
2 #s name "tiny03"
3 #s desc "Instance with large MAF."
4 #p 4 8
5 (( (1,2), (3,4)), ((5,6), (7,8))) ;
6 (7, (3, (2, (5, (4, (6, (1,8))))))) ;
7 (( (1,6), (3,8)), ((5,2), (7,4))) ;
8 (( (1,7), (2,8)), ((5,3), (6,4))) ;
9 #x treedecomp [4, [[8,11,12,14], [6,8,11,12,14], [6,8,11,12,32], [6,8,11,22,32], [6,11,22,25,32], [1,11,22,25,32], [6,8,12,29,32], [6,12,29,36], [4,12,20,29,36], [8,15,29,
  ↳ 32], [7,15,16,29,32], [6,11,12,14,28], [8,11,28,33], [2,11,18,28,33], [8,12,14,28,35], [8,12,26,35], [3,12,17,26,35], [5,14,19,28,35], [10,11,12], [9,10,13], [23],
  ↳ [24], [27], [30], [31], [34]], [[1,2], [1,13], [1,20], [1,22], [1,23], [1,24], [1,25], [1,26], [1,27], [2,3], [2,12], [3,4], [3,7], [4,5], [5,6], [7,8], [7,10], [8,9], [10,11], [13],
  ↳ 14], [13,16], [14,15], [16,17], [16,19], [17,18], [20,21]]]

```

Feasible solution:

1	1;
2	3;
3	4;
4	(5,7);
5	6;
6	8;
7	2;



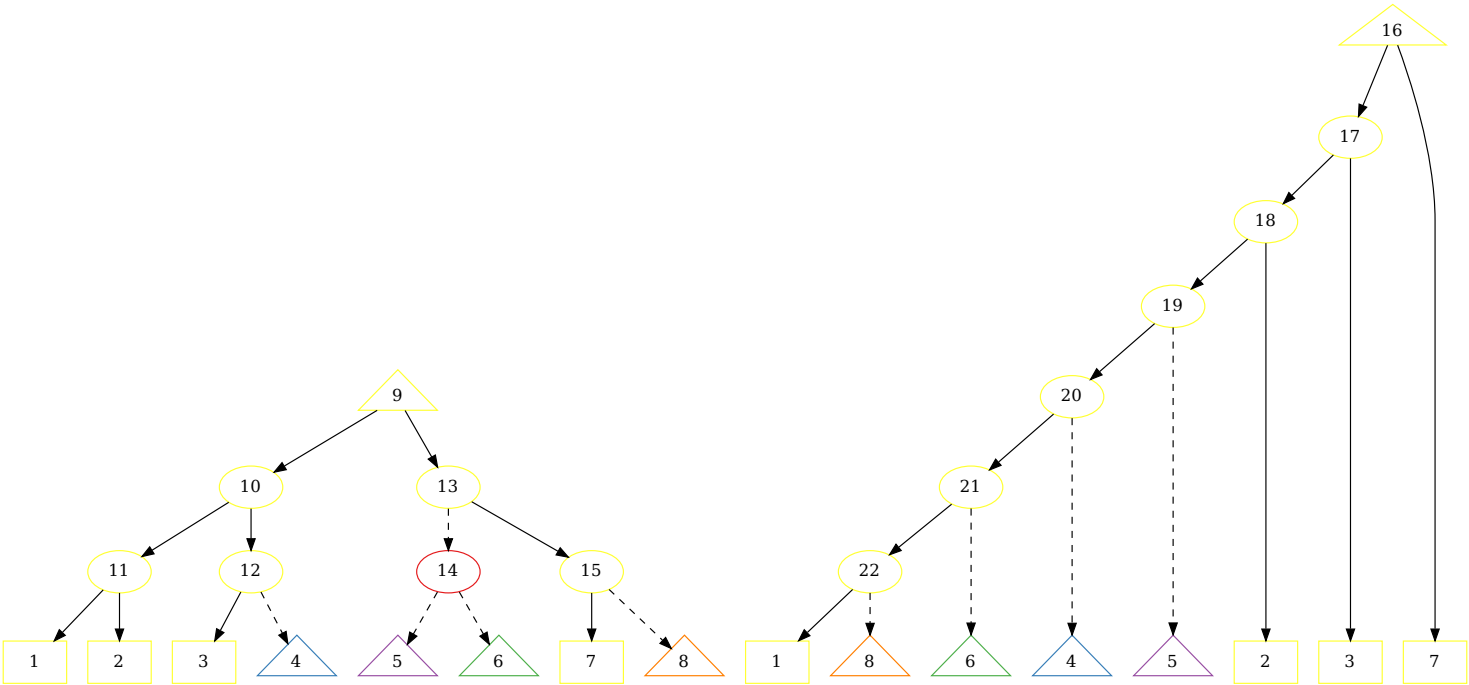
tiny04.nw

Instance:

```
#s hash "0x2bc29507df4c3762ff0726"
#s name "tiny04"
#s desc "Instance where Cherry-Rule is not applicable"
#p 2 8
(((1,2),(3,4)),((5,6),(7,8)));
(7,(3,(2,(5,(4,(6,(1,8)))))));
#x treedecomp [2,[[10,13],[10,11,13],[11,13,22],[1,11,22],[13,15,22],[7,15,16],[8,15,22],[2,11,18],[10,12],[3,12,17],[4,12,20],[13,14],[5,14,19],[6,14,21],[9,10,13]],[[1,2],[1,9],[1,12],[1,15],[2,3],[2,8],[3,4],[3,5],[5,6],[5,7],[9,10],[9,11],[12,13],[12,14]]]
```

Feasible solution:

```
4;
6;
5;
8;
(((1,2),3),7);
```



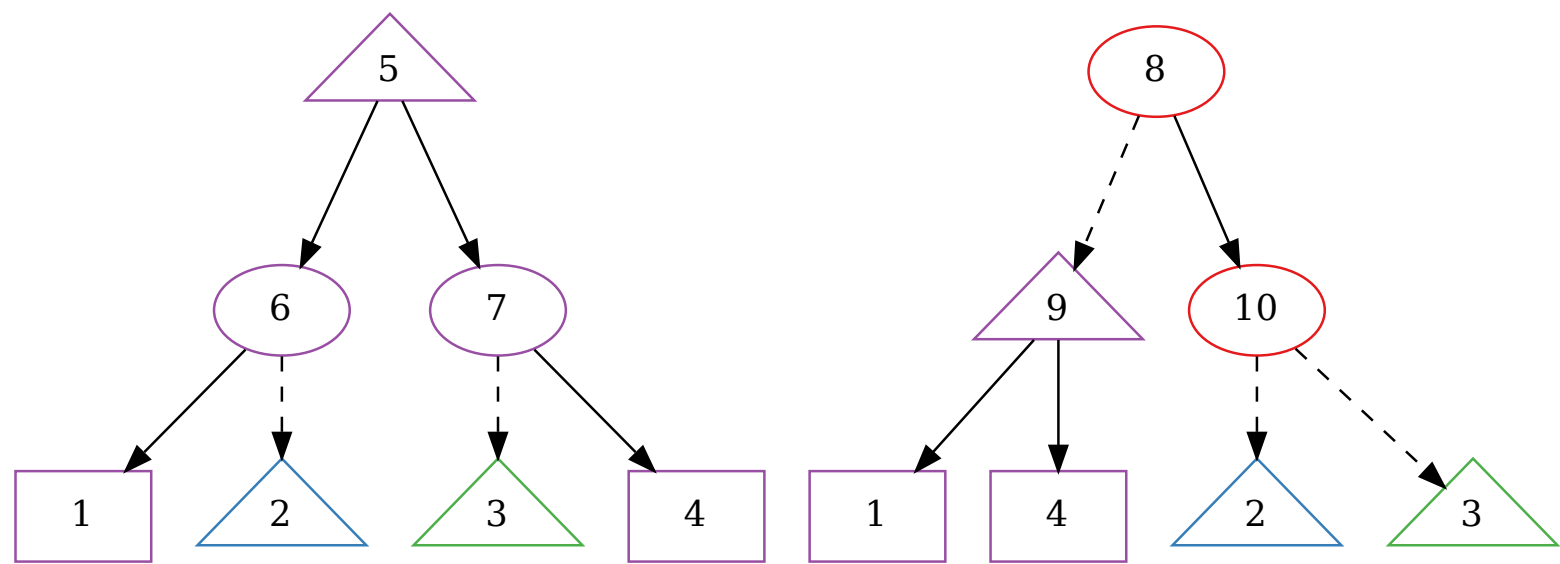
tiny05.nw

Instance:

```
1 #s hash "0x100e2f0731d8e7d508f896"  
2 #s name "tiny05"  
3 #p 2 4  
4 ((1,2),(3,4));  
5 ((1,4),(2,3));  
6 #x treedecomp [2,[[6,7],[6,7,9],[1,6,9],[4,7,9],[6,7,10],[2,6,10],[3,7,10],[5,6,7],[8]],[[1,2],[1,5],[1,8],[1,9],[2,3],[2,4],[5,6],[5,7]]]
```

Feasible solution:

```
1 2;  
2 3;  
3 (1,4);
```



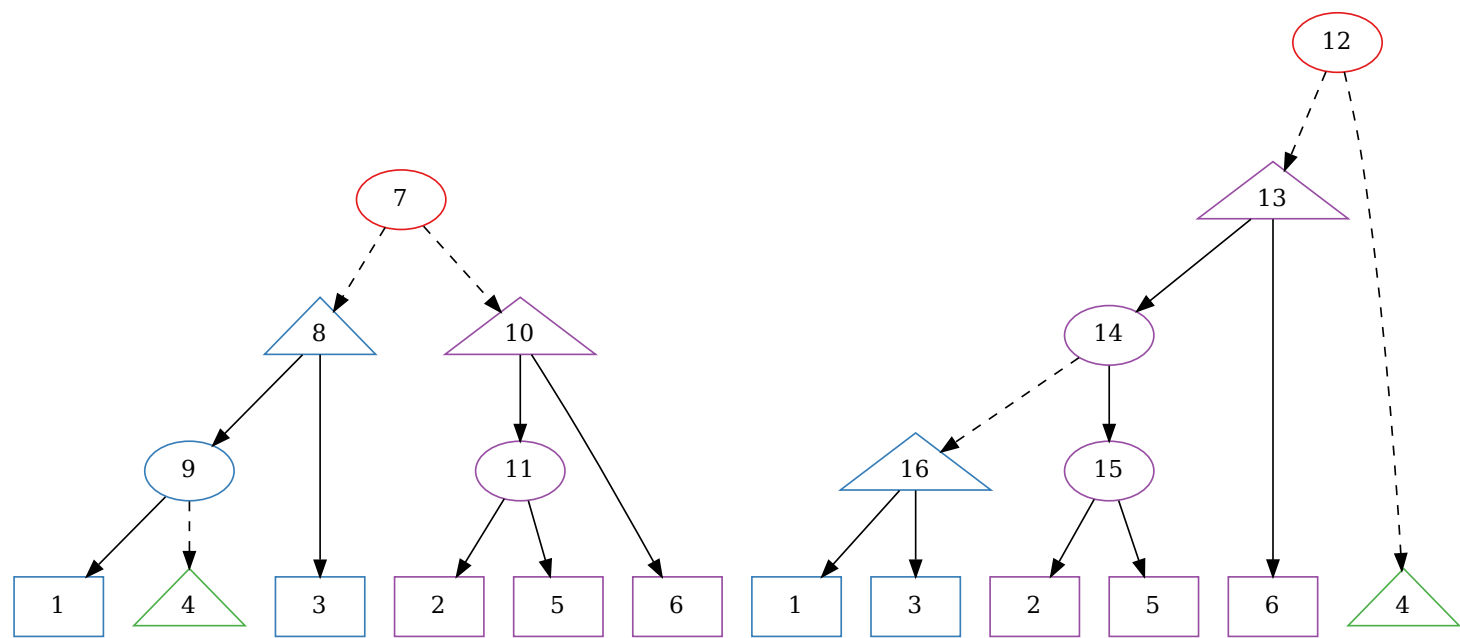
tiny06.nw

Instance:

```
#s hash "0x19bd9de48e72ecf85b7454"
#s name "tiny06"
#p 2 6
((3,(4,1)),((2,5),6));
(((2,5),(3,1)),6),4);
#x treedecomp [2,[ [7],[7,8],[8,9,16],[1,9,16],[3,8,16],[4,9,12],[7,10,11],[11,15],[2,11,15],[5,11,15],[6,10,13],[14]],[ [1,2],[1,7],[1,12],[2,3],[3,4],[3,5],[3,6] ],
  ↪ [7,8],[7,11],[8,9],[8,10]]]
```

Feasible solution:

```
(3,1);
4;
((2,5),6);
```



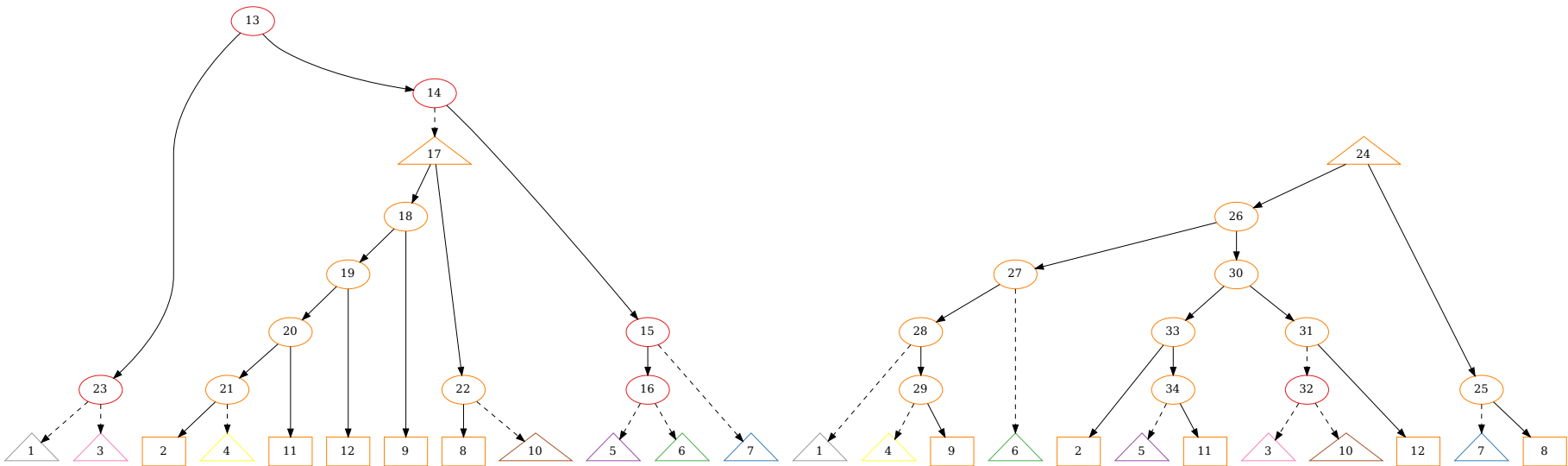
tiny07.nw

Instance:

```
#s hash "0x2ccaa5227a4405ec0d71ac"
#s name "tiny07"
#p 2 12
# derived from https://www.plantplus.cn/treehub/tree/T00004945
(((6,5),7),((((4,2),11),12),9),(8,10))), (1,3));
# derived from https://www.plantplus.cn/treehub/tree/T00123573
((8,7),(((1,(9,4)),6),(((3,10),12),((5,11),2))));
#x treedecomp
  ↳ [3, [[14,22,34], [14,22,23], [1,23,28], [22,23,32], [3,23,32], [10,22,32], [13,14,23], [14,18,22,34], [18,20,21,34], [2,21,33], [18,21,29], [4,21,29], [9,18,29], [11,20,34],
  ↳ [18,19,20], [12,19,31], [14,17,18,22], [14,16,22,34], [5,16,34], [6,16,27], [14,15,16,22], [15,22,25], [7,15,25], [8,22,25], [24], [26], [30]], [[1,2], [1,8], [1,18], [1,25],
  ↳ [1,26], [1,27], [2,3], [2,4], [2,7], [4,5], [4,6], [8,9], [8,17], [9,10], [9,11], [9,14], [9,15], [11,12], [11,13], [15,16], [18,19], [18,20], [18,21], [21,22], [22,23], [22,24]]]
```

Feasible solution:

```
7;
6;
5;
((((11,2),12),9),8);
4;
10;
3;
1;
```



tiny08.nw

Instance:

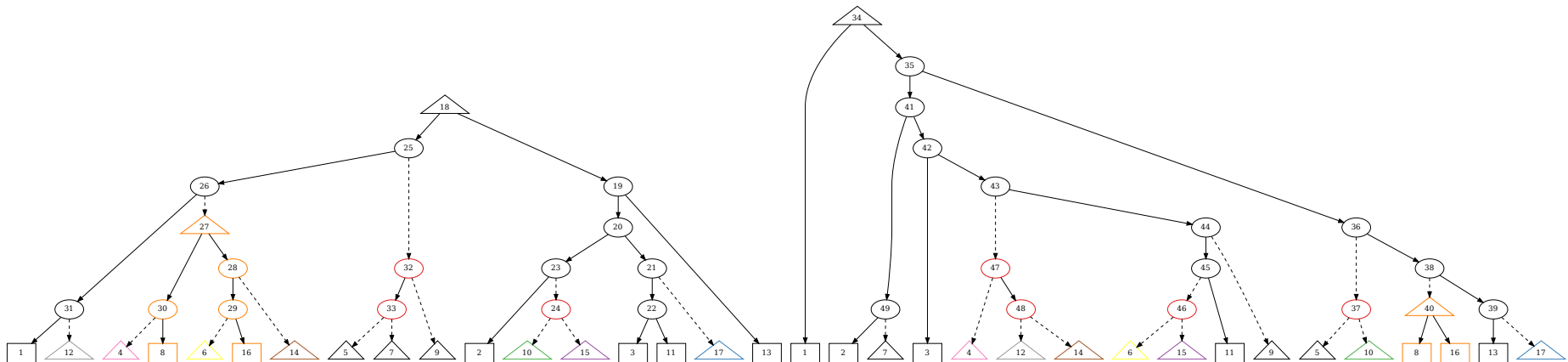
```
1 #s hash "0x2b34f682a195a6438ac412"
2 #s name "tiny08"
3 #p 2 17
4 # derived from https://www.plantplus.cn/treehub/tree/T00122092
5 (((((3,11),17),(2,(15,10))),13),((((16,6),14),(4,8)),(1,12)),(9,(5,7))));
6 # derived from https://www.plantplus.cn/treehub/tree/T00122188
7 (((5,10),((17,13),(16,8))),((3,((9,(11,(15,6))),4,(14,12))),(2,7))),1);
8 #x treedecomp [3,[2,24,25],[24,25,27,31],[1,31,34],[24,27,29,31],[27,29,30],[4,30,47],[29,30,40],[8,30,40],[16,29,40],[24,29,46],[6,29,46],[15,24,46],[27,29,31,
↳ 48],[12,31,48],[27,28,29,48],[14,28,48],[25,26,27,31],[2,20,24,25],[19,20,21,25],[21,22],[3,22,42],[11,22,45],[19,21,39],[13,19,39],[17,21,39],[18,19,25],[2,20,
↳ 23,24],[2,24,25,33],[24,33,37],[5,33,37],[10,24,37],[2,7,33,49],[25,32,33],[9,32,44],[35],[36],[38],[41],[43]],[[1,2],[1,18],[1,28],[1,35],[1,36],[1,37],[1,38],
↳ [1,39],[2,3],[2,4],[2,17],[4,5],[4,10],[4,13],[5,6],[5,7],[7,8],[7,9],[10,11],[10,12],[13,14],[13,15],[15,16],[18,19],[18,27],[19,20],[19,23],[19,26],[20,21],
↳ [20,22],[23,24],[23,25],[28,29],[28,32],[28,33],[29,30],[29,31],[33,34]]]
```

Feasible solution:

```

1  17;
2  10;
3  15;
4  (16,8);
5  6;
6  14;
7  4;
8  12;
9  9;
10 5;
11 7;
12 (((3,11),2),13),1);

```



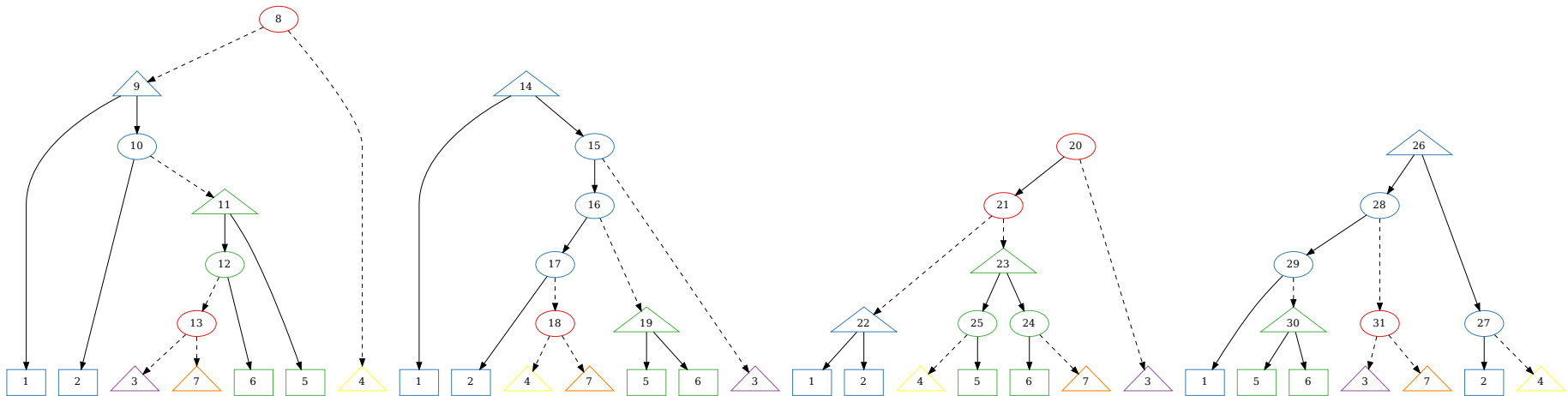
tiny09.nw

Instance:

```
#s hash "0x21fb78dd8c1a2e1f2c8502"
#s name "tiny09"
#p 4 7
# derived from https://www.plantplus.cn/treehub/tree/T00000561
((((((3,7),6),5),2),1),4);
# derived from https://www.plantplus.cn/treehub/tree/T00000676
((((7,4),2),(5,6)),3),1);
# derived from https://www.plantplus.cn/treehub/tree/T00002688
(((2,1),((7,6),(5,4))),3);
# derived from https://www.plantplus.cn/treehub/tree/T00003788
((2,4),(((6,5),1),(7,3)));
#x treedecomp [3,[7,10,25],[4,7,10,25],[4,9,10,22],[1,9,22],[1,14],[1,29],[4,10,22,27],[2,10,22,27],[2,17],[4,8,9],[4,7,18],[7,13,31],[3,13,31],[3,15],[3,20],[6,7,10,25],[6,10,11,25],[5,6,11,25],[5,6,19],[5,6,30],[6,12],[6,7,24],[16],[21],[23],[26],[28]],[[1,2],[1,12],[1,16],[1,23],[1,24],[1,25],[1,26],[1,27],[2,3],[2,11],[3,4],[3,7],[3,10],[4,5],[4,6],[7,8],[8,9],[12,13],[13,14],[13,15],[16,17],[16,21],[16,22],[17,18],[18,19],[18,20]]]
```

Feasible solution:

(1, 2);
(5, 6);
3;
7;
4;



tiny10.nw

Instance:

```
#s hash "0x2db593400e7dc786c2f45b"
#s name "tiny10"
#p 2 11
(1,(2,(3,(4,(5,(6,(7,(8,(9,(10,11)))))))));
(11,(10,(9,(8,(7,(6,(5,(4,(3,(2,1)))))))));
#x treedecomp
  ↳ [2,[[17],[15,16,17],[13,14,15],[12,13,31],[1,12,31],[2,13,31],[3,14,30],[4,15,29],[5,16,28],[6,17,27],[17,18],[7,18,26],[18,19],[8,19,25],[19,20],[9,20,24],[20,
  ↳ 21],[10,21,23],[11,21,22]],[[1,2],[1,10],[1,11],[2,3],[2,8],[2,9],[3,4],[3,7],[4,5],[4,6],[11,12],[11,13],[13,14],[13,15],[15,16],[15,17],[17,18],[17,19]]]
```

Feasible solution:

```
(3,4);
(5,6);
(7,8);
(10,11);
9;
(1,2);
```

